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TOPICAL REVIEW

Visual Language Models: Foundations and Applications

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ABSTRACT Visual Language Models (VLMs) have swiftly accelerated the blending of the visual modality with textual information, enabling more natural and contextually aware human–AI interaction. This survey offers an in-depth overview of foundations, capabilities and constraints of vision–language systems, covering (i) *classical VLMs* focused on perception, retrieval and grounding, (ii) *LLM-grounded multimodal large language models (MLLMs)* for open-ended generation and instruction following and (iii) emerging *agentic/tool-augmented* multimodal systems. We introduce a taxonomy of architectures, training objectives and benchmark data, commenting on how design decisions impact performance and deployability. We then discuss evaluation processes, balancing capability measures with reliability measures like robustness, fairness and efficiency, and we include an illustrative “complete” evaluation report card. Applications across fields like healthcare, education, scientific discovery and creative arts are discussed to illuminate both potential and practical challenges. Finally, we summarize open issues in scalability, explainability and trustworthy operation. The literature covered in this survey spans representative works published up to December 31, 2025.

INDEX TERMS Visual language models, multimodal learning, foundation models, vision and language, machine learning.

I. INTRODUCTION

Visual Language Models (VLMs) are a new AI innovation because they enable computers to simultaneously reason and process both visual information and text information. This assistance in a multimodal format provides the foundation upon which the tasks of grounded understanding such as cross-modal retrieval and embodied vision-language navigation, Visual Question Answering (VQA) and image captioning arise. VLMs bridge the semantic gap between vision and language and thus provide the computational

platform upon which human-computer interaction becomes more natural and more context-sensitive [1].

In this survey we use *VLM* as an umbrella term, but we explicitly distinguish: (i) *classical VLMs* (perception/retrieval/grounding), (ii) *MLLMs* that connect a vision encoder to an LLM-style generator for open-ended instruction following and (iii) *agentic multimodal systems* that integrate external tools (retrievers, knowledge graphs, planners, OCR and memory). Our emphasis is on architectural and training foundations that underlie all three; tool ecosystems are discussed as deployment-oriented extensions rather than exhaustively cataloged. The covered literature includes representative works published up to December 31, 2025.

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TABLE 1. List of abbreviations.

Abbreviation	Definition
VLM	Visual Language Model
CV	Computer Vision
NLP	Natural Language Processing
MLP	Multilayer Perceptron
CNN	Convolutional Neural Network
ViT	Vision Transformer
LLM	Large Language Model
MMT	Multimodal Transformer
VQA	Visual Question Answering
CLIP	Contrastive Language–Image Pretraining
ALIGN	A Large-scale Image and Noisy-text embedding model
BLIP	Bootstrapping Language-Image Pretraining
CoT	Chain of Thought
PEFT	Parameter-Efficient Fine-Tuning
RLHF	Reinforcement Learning with Human Feedback
FID	Fréchet Inception Distance
BLEU	Bilingual Evaluation Understudy (text generation metric)
CIDEr	Consensus-based Image Description Evaluation (captioning metric)

Their surge in prominence has been mostly due to the use of transformer-based architectures, which have high attention mechanisms capable of modeling long-range relationships across and within modalities. The earlier models were all about closely coupled textual and visual inputs and region-based features or object detectors. But the subfield has developed very quickly and has led to scalable and more integrated models with end-to-end raw images and text tokens processing. Dual-encoder methods like CLIP and ALIGN have been astounding in their success in zero-shot transfer and contrastive pretraining, while fusion-based and encoder-decoder methods have established the state of the art in multimodal reasoning and generative tasks [2]. Because “zero-shot” results may reflect both genuine semantic generalization *and* unintended test-set overlap with web-scale pretraining corpora, we emphasize the importance of data de-duplication and contamination audits when interpreting reported zero-shot performance. Because “zero-shot” results may reflect both genuine semantic generalization *and* unintended test-set overlap with web-scale pretraining corpora, we emphasize the importance of data de-duplication and contamination audits when interpreting reported zero-shot performance.

In addition, combining LLMs with vision has also opened up new frontiers, allowing general-purpose systems to execute multi-aspect multimodal commands and do open-ended reasoning. These advancements saw the coming together of vision-language with other Foundation Model trends of instruction tuning and few-shot learning [3].

In order to illustrate the typical pipeline of VLMs, Figure 1 provides an exemplary schematic architecture. A VLM typically takes multimodal inputs like images and written input, encodes them with custom encoders and alignment layers and combines the extracted representations with a multimodal core. Decoded integrated features then yield outputs like captions, visual question responses or actions corresponding to a given task. Though the figure hides most of the architectural variations, it reveals the typical

building blocks (i.e. vision and language encoders, alignment modules and multimodal integration) of the larger number of contemporary VLMs.

Several surveys have already reviewed aspects of multimodal learning and VLMs. Baltrušaitis et al.’s influential paper [4] provided us with one of the first thorough treatments of multimodal machine learning and presented a taxonomy centered on representation, translation, alignment and fusion. Although enormously influential, the survey comes before the age of large pretrained models and does not reflect the paradigm shift VLMs of the present time have provided.

Much later, Zhang et al. [5] released a comprehensive survey of VLMs for vision-focused tasks. Their comparison was performed systematically over architectures and training procedures applied to object detection, segmentation and their duals. Their focus, however, remained largely restricted to vision tasks and was furthermore not extended over more general uses like visual question answering, multimodal reasoning or generation abilities.

At the same time, a line of milestone models and benchmarks have driven the field. LXMERT [6] and BLIP [7] set transformer-based cross-modal encoders and unified pretraining techniques in motion and ALIGN [8] demonstrated the power of noisy large-scale image–text supervision. Benchmarks such as OK-VQA [9] highlighted external knowledge in visual question answering and more recent systems such as MiniGPT-4 [10] exhibited the combination of large language models and visual encoders. These collectively portray the rapid rise of VLMs across diverse application spaces. Despite such progress, recent surveys and model-focused reviews fail to address a series of gaps.

Firstly, all existing research treats VLMs either as an extension of computer vision or an extension of NLP, never explaining how they possibly comprise a unifying paradigm. Second, there is no single comprehensive taxonomy found in the literature to place architectures, training schemes and test suites within a single unifying framework. Third, evaluation schemes are still fragmented: some undertaking comparisons of data set, others in-task measures, but few attempting a joined-up view across evaluation axes. Finally, ethical and interpretability issues, which are critical in moving to deployment in the wild, are almost invariably discussed only tangentially. In our survey here, we attempt to fill these gaps with (i) placing VLMs in the broader history of multimodal AI, (ii) presenting an integrated taxonomy of architecture, training plans and test collections, (iii) cross-testing test suites across tasks and (iv) contrastingly critiquing open scalability, robustness, interpretability and ethical issues.

Compared with prior surveys, our perspective is explicitly *deployment-facing*: we connect architectures and objectives to supervision regimes, evaluation/reporting practices (including robustness and efficiency) and application KPIs, providing cross-walk tables and checklists intended to support model selection and reproducible reporting.

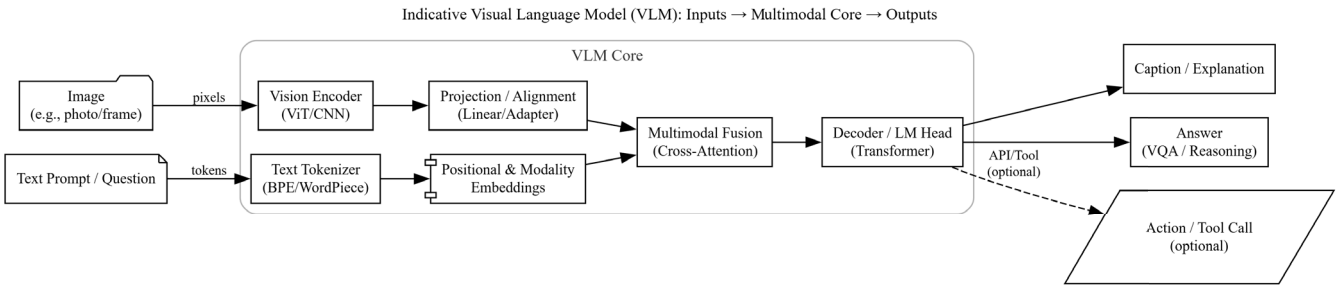


FIGURE 1. Indicative architecture of a Visual Language Model (VLM), showing the flow from multimodal inputs through the core fusion layers to diverse outputs.

TABLE 2. VLM design space at a glance (visual summary of the main architectural and training choices).

Design Axis	Typical Options	What it optimizes	Practical implication (when to choose)
Architecture	Dual-encoder; fusion/cross-attention; encoder–decoder; one-tower	Retrieval vs reasoning vs generation	Dual-encoders are efficient and strong at zero-shot retrieval; fusion/enc–dec/one-tower excel at grounded reasoning and generation but are heavier to train/infer.
Tokenization of vision	Regions; patches (ViT); pixels+OCR tokens	Fine-grained grounding vs simplicity	Regions help localized grounding (text-in-image, VQA); patch tokenization scales cleanly; OCR tokens are critical for documents/scenes with text.
Objective mix	Contrastive; ITM/MLM; autoregressive (AR) generation; instruction tuning	Alignment vs controllability	Contrastive gives broad representation and robust retrieval; ITM/MLM improve cross-modal matching; AR/instruction enables open-ended dialogue and tool-like behavior.
Data regime	Curated captions/Q&A; web alt-text; bootstrapped web; domain corpora	Label quality vs scale	Curated data improves grounding but is small; web-scale boosts generalization but needs filtering; domain corpora demand domain metrics and careful safety checks.
Deployment target	Edge/real-time; safety-critical	server; Latency/cost vs reliability	For edge: prefer dual-encoder or lightweight bridging + PEFT/quantization; for safety-critical: add calibration, abstention, retrieval grounding and robust evaluation.

The contributions of the paper can be summarized as follows:

- Taxonomy of VLMs: We organize vision–language models based on architecture, learning targets and data and justify design compromises.
- Evaluation framework: We integrate capability-oriented measures with reliability attributes such as robustness, efficiency and fairness and relate them with representative benchmarks.
- Applications and limitations: We describe existing VLM applications across eight application domains, presenting accomplishments and recurring chokepoints.
- Directions for future: There are identifiable open issues in scalability, interpretability and ethical implementation and research and practice directions are sketched out.

Based on this motivation and groundwork, the rest of this paper is organized as follows. Section II provides an overview of the historical development and technical roots of Visual–Language Models, tracing the progression from initial captioning pipelines to multimodal foundation stacks.

Section III establishes a unifying taxonomy of architectures, objectives, interaction methods and input granularities. Section IV presents advances in architectures beyond core families, while Section V discusses training frameworks, optimization techniques and dataset regimes. Section VI discusses evaluation criteria, standards and robustness axes. Section VII canvasses representative application areas, before Section VIII condenses open challenges and opportunities for research. Finally, Section IX concludes with commentary on how VLMs will shape multimodal AI in the years to come.

II. BACKGROUND AND FOUNDATIONS

The modern trajectory of image captioning research, culminating in large-scale vision–language models (VLMs), has progressed from convolutional neural network (CNN)-based encoders to recurrent neural network (RNN)-based captioners (e.g., “Show and Tell”, “Show, Attend and Tell”), [11], [12], [13] and more recently to transformer-based, pretraining-intensive systems that align and fuse visual and textual modalities at scale. Table 3 summarizes key milestones: cross-modal transformers (LXMERT) [6],

TABLE 3. Timeline of key milestones in Visual Language Models (VLMs).

Year	Model / Benchmark	Contribution / Significance
2018	Multimodal Survey [4]	Early comprehensive taxonomy of multimodal learning (representation, translation, alignment, fusion).
2019	LXMERT [6]	Introduced cross-modal transformers for VQA and captioning; established encoder-decoder paradigm.
2019	OK-VQA [9]	Benchmark requiring external knowledge for visual question answering; emphasized reasoning beyond perception.
2021	ALIGN [8]	Demonstrated large-scale contrastive training with noisy image-text pairs, achieving strong generalization.
2022	BLIP [7]	Unified vision-language pretraining and generation; enabled zero-shot and fine-tuned performance across tasks.
2023	MiniGPT-4 [10]	Integrated large language models with visual encoders; demonstrated conversational and generative multimodal capabilities.
2024	Survey of VLMs for Vision Tasks [5]	Comprehensive review of vision-task-oriented VLM applications; detailed architectures and training strategies.
2025	Emerging VLM Applications [14]–[16]	Expanded use cases including multimodal misinformation detection, security risks and domain-specific VQA.

web-scale contrastive pretraining (ALIGN) [8], unified pretraining for understanding/generation (BLIP) [7] and Large Language Model (LLM)-grounded stacks (MiniGPT-4) [10]. This history of the field encourages these axes of design explored in this overview: (i) component-level building blocks, (ii) training paradigms/targets and (iii) the data and measures governing them.

A. CORE BUILDING BLOCKS

Vision encoders take out visual tokens through transformer or convolutional backbones, with object-centric or patch/grid features passing through downstream fusion [6]. Lightweight architectures like ViLT work directly from patch embeddings with reduced visual pre-processing [17].

Text encoders and decoders follow several patterns. Dual encoders learn joint spaces with a separate text tower (ALIGN) [8], while fusion encoders use bidirectional transformers for Image-Text Matching (ITM)/Masked Language Modeling (MLM) [6], [17]. Decoder-based stacks instead pair a vision encoder with an LLM to generate open-ended text (Flamingo, MiniGPT-4) [10], [18].

Projection and alignment layers cast modalities onto joint spaces, either contrastively [8] or to generate via inputting decoder token spaces [10]. Cross-modal fusion realizes token-level interaction via cross-attention between visual and textual streams (LXMERT) [6], interleaved attention inside an LLM (Flamingo) [18] or through a single multimodal transformer (ViLT) [17].

Finally, heads of tasks and interfaces support application-level outputs: similarity heads support retrieval and zero-shot transfer and decoder-based interfaces drive captioning and QA (BLIP, Vid2Seq) [7], [19].

B. TRAINING PARADIGMS (BIRD’S-EYE VIEW)

Three traditional frameworks define VLM training today. The *Two-Tower* approach employs dual encoders with contrastive alignment, enabling scalable retrieval and zero-shot transfer [8], [20]. The *Two-Leg* design introduces dual streams with cross-attention, which supports stronger

grounding and reasoning through ITM/MLM and generative objectives [6], [7], [18]. Finally, the *One-Tower* paradigm uses a single multimodal transformer over patches and tokens, favoring simplicity and efficiency [17]. LLM-backed variants extend these paradigms by grounding pretrained decoders with visual tokens for instruction following [10] and [18]. A comparison is provided side-by-side in Table 4.

C. DATA REGIMES THAT SHAPE OBJECTIVES

The *supervision regime* largely determines which pretraining *objectives* are feasible and effective (see Table 5). Instruction-rich or pre-curated data sets generally accept *generative*, Hybrid or bootstrap sources tend to be used in *multitask mixtures with light adapters*. Domain-specific datasets usually mandate *adapters* and *domain-aware evaluation* (Sec. VI).

D. EVALUATION AT A GLANCE

Captioning uses BLEU/METEOR/ROUGE/CIDEr [28], [29], [30], [31]; retrieval relies on Recall@k/MRR/nDCG [32], [33], [34]; VQA reports accuracy [22]. Human judgments, pairwise preferences and domain-expert agreement remain essential for open-ended generation and safety-critical settings. We summarize metrics in Table 7 and complement them with robustness dimensions (Table 8).

E. ILLUSTRATIVE EXAMPLE: REPORTING A “COMPLETE” EVALUATION PROTOCOL

To increase the practical value of the survey, Table 6 provides a concise example of how authors can report (i) *core task metrics*, (ii) *robustness checks* and (iii) *computational considerations* in a single “evaluation card”. Values are intentionally illustrative (placeholders) and should be replaced by results obtained under the paper’s stated data splits, prompts and hardware.

F. POSITIONING WITHIN EXISTING SURVEYS

Prior surveys cover multimodal taxonomies [4], Visual Language Pretraining (VLP) overviews [35], [36], [37],

TABLE 4. VLM pretraining frameworks at a glance: architectural sketch, objectives, trade-offs and representative models.

Framework	Architectural Sketch	Typical Objectives	Pros	Cons	Representative Models
Two-Tower (Dual Encoder / Contrastive)	Separate vision & text encoders; align pooled embeddings in a shared space	Contrastive (InfoNCE), image↔text retrieval	Scales to web data; efficient retrieval/zero-shot; simple deployment	Limited token-level grounding/reasoning; fusion is implicit	ALIGN [8]
Two-Leg (Dual Stream with Fusion)	Visual & textual streams interact via cross-attention (bi- or interleaved)	ITM/MLM; generative (QA/caption) when paired with a decoder	Fine-grained alignment; strong VQA/reasoning; flexible supervision	Heavier compute/memory; more complex training	LXMERT [6], Flamingo [18], BLIP [7]
One-Tower (Single Multimodal Transformer)	One transformer consumes visual patches + word tokens (type/positional embeddings)	ITM/MLM (with/without auxiliary signals); light generative heads	Simple stack; fewer params; faster train/infer; easy scaling	May lag on fine-grained grounding without regions/OCR	ViLT [17]

TABLE 5. Supervision regimes, primary objectives and representative data sources (with examples).

Supervision Regime	Primary Objectives	Representative Datasets / Sources	Notes (label quality, scale, typical use)
Curated captions / Q&A	ITM, MLM; AR generation	COCO [21]; VQA v2 [22]; TextVQA [23]	Clean labels; token-level alignment; strong for captioning/VQA and grounding.
Web-scale alt-text (weak)	Contrastive (InfoNCE)	Conceptual Captions [24]; LAION-5B [20]	Massive scale offsets noise; excels at retrieval & zero-shot classification (CLIP/ALIGN [8], [25]).
Bootstrapped cleaned web	Mixed (contrastive + ITM/MLM + AR)	BLIP bootstrapping [7]	Model-in-the-loop rewriting/filters upgrade noisy pairs; supports unified training.
Domain-specific corpora	ITM/MLM + adapters; selective AR	Medical/RS/Industrial sets (e.g., CLIP-MoA [26])	Distribution shift; need adapters and domain metrics; validate with experts.
Instruction-style corpora	AR instruction-following; RLHF-friendly	Task/prompt collections; domain instruction sets (e.g., Diagnostics-LLaVA [27])	Enables open-ended dialogue and tool use; report EM/F1 + human prefs.
Video/narration	AR dense captioning; alignment	Narrated videos; Vid2Seq pretraining [19]	Pseudo-labels from speech; temporal grounding for events.

multimodal transformers [38], efficiency-oriented VLM surveys [39] and VLMs for vision-centric tasks [5]. Table 9 contrasts their emphases with ours. Our contribution is a unified, practice-oriented synthesis that ties *architectures* and *objectives* to *data regimes*, *metrics/KPIs* and *applications*, with comparative tables to aid design decisions.

We reserve *unified* (generalist) VLMs to describe single parameterizations with many I/O formats and tasks via common backbones and blending objectives. For clarity, throughout this survey we distinguish three (often overlapping) system types:

- Classical VLMs: models primarily designed for *perception*, *retrieval* and *grounding* (e.g., cross-modal retrieval, ITM/MLM-based matching, region/patch-level alignment, VQA and captioning with explicit visual–textual grounding).
- Multimodal large language models (MLLMs): systems that couple a visual encoder with an LLM-style decoder (or tightly integrated multimodal transformer) to support *open-ended generation* and *instruction following*, typically producing free-form text explanations, multi-turn dialogue and long-form reasoning.

- Agentic multimodal systems: MLLM/VLM backbones augmented with *external tools and memory* (e.g., retrieval modules, knowledge graphs, OCR, planners and execution engines) so that outputs are grounded not only in the input image(s) but also in dynamically queried evidence and actionable tool results.

In scope, we review the architectural foundations of classical VLMs and the bridging mechanisms that lead to MLLMs, and we summarize emerging agentic patterns as a deployment-oriented extension (rather than an exhaustive survey of tool ecosystems). Recent agentic collaborative reasoning over heterogeneous knowledge sources (e.g., robust text-to-SPARQL over knowledge graphs) exemplifies the tool-augmented trend [40], while distributed/federated adaptation of large models is becoming increasingly relevant for privacy- and bandwidth-constrained deployment settings [41].

Exemplars include Flamingo (interleaving cross-attention into an LLM for few-shot multimodality) [18], BLIP/BLIP-2 (bridging contrastive, ITM/MLM and generative pretraining; Q-Former for efficient LLM interfacing) [7], [42] and

TABLE 6. Example of a complete evaluation report card for a VLM/MLLM on a representative benchmark suite.

Evaluation Block	What to report	Example fields	Notes (recommended practice)
Task setup	Dataset(s), split, prompt template, decoding, temperature, max tokens	VQA v2 (val); prompt v3; greedy; $T = 0$; 128 tokens	Fix prompts/decoding for comparability; report any test-time augmentation/tool use.
Core metrics	Primary score(s) + confidence intervals	VQA acc.=78.2% (± 0.3); R@1=62.0	Use bootstrap CIs where feasible; include at least one retrieval and one generation metric when relevant.
Calibration / abstention	ECE/Brier; accuracy@coverage; selective refusal rate	ECE=0.08; Acc@90% cov.=82.1%	Critical for safety-critical domains; define abstention policy and thresholding.
Robustness: domain shift	In-domain vs OOD scores; relative drop	OOD acc.=69.4% (-11.3 pts)	Specify the OOD set (medical, RS, docs, etc.) and report subgroup gaps.
Robustness: input corruption	Sensitivity to blur/occlusion/noise; missing modality tests	Blur-5 drop: -4.2 pts; no-text drop: -6.0 pts	Include at least one vision corruption suite and one text/prompt perturbation suite.
Robustness: adversarial / prompt attacks	Jailbreak/prompt-injection success; image-trigger/backdoor checks (if applicable)	Injection success: 7% (lower is better)	Report threat model and filtering; avoid security theater by stating exact protocol.
Efficiency	Latency, throughput, peak VRAM, energy proxy	410 ms/img; 2.4 img/s; 18 GB	Report batch size, precision (fp16/bf16/int8) and I/O; include cold-start vs warm.
Training/finetuning cost	Train tokens/steps; GPU-hours; FLOPs (if available)	8 A100 $\times$ 12 h; 4.2e20 FLOPs	Separate pretraining from finetuning; provide reproducibility details.
Ablations	Objective mix, adapter choice, data filtering	w/o ITM: -2.1 pts; LoRA vs full: $\approx$	Tie to claims: each major design claim should have at least one supporting ablation.

MiniGPT-4 (vision encoder to LLM decoder for instruction following) [10]. These systems reduce heads with task specificity but increase capability with in-context learning and instruction tuning. In the following, we construct architectural and objective taxonomies and go back to deployment-facing consequences with datasets, evaluation and uses.

Section II provides only the essentials for the rest of the paper. Detailed analyses of *architectures* and *training objectives* follow next, while *datasets*, *evaluation* and *applications* are covered in Sections V–VII.

III. TAXONOMY OF VISUAL LANGUAGE MODELS

VLMs vary along several largely orthogonal axes: (i) *architectural family*, (ii) *training objectives*, (iii) *modality interaction strategy* and (iv) *input granularity & perceptual front-ends*.

We distinguish *structural*, *long-lasting* criteria—architecture family (e.g., dual-encoder vs. fusion vs. encoder-decoder), interaction mechanism (alignment vs. cross-attention) and objective family (contrastive vs. ITM/MLM vs. autoregressive/instruction)—from *stage-specific* criteria that may shift with the research landscape, such as popular benchmark suites, instruction formats and tool/agent integration patterns. The taxonomy (Fig. 2) is therefore organized around the stable axes, while stage-specific trends are discussed as evolving layers on top of these foundations (Secs. VI–VII). Figure 2 offers a comprehensive view of the VLMs’ domain.

Several axes define the taxonomy of VLMs. The first is the *architectural family*, which spans *Two-Tower* dual encoders with contrastive alignment, *Two-Leg* dual streams with cross-attention, *One-Tower* multimodal transformers and *Encoder-Decoder* stacks grounded on LLMs (Section II-B, Table 4).

A second axis concerns the *training objectives*, including contrastive alignment, ITM/MLM fusion losses and generative or instruction-tuned objectives (Section III-B).

The *interaction strategy* provides a third axis, distinguishing early or late fusion, cross-attention mechanisms and retrieval-only alignment (Section III-C).

Finally, *input granularity* ranges from global image-level features to region-level detector outputs, patch-level tokens (ViT) and Optical Character Recognition (OCR)/text tokens when required (Section III-D).

A. ARCHITECTURAL FAMILIES

VLM architectural designs belong to four major families. The *Two-Tower* method uses two image and text encoders with contrastively aligned pooled embeddings more closely aligned with retrieval and zero-shot transfer [8], [25]. The *Two-Leg* family instead uses dual streams with cross-attention, allowing token-level grounding and strong performance on VQA [6], [7], [18]. Another option is the *One-Tower* architecture, where a single multimodal transformer receives patch and word tokens with a bias towards efficiency and lightness [17]. Finally, *Encoder-Decoder* stacks project visual features into a decoder-only LLM, enabling instruction-following and open-ended generation [10]. Table 10 depicts a compact comparison across representative families including unified/hybrid approaches.

B. PRETRAINING OBJECTIVES (BY FAMILY)

Pretraining objectives for VLMs fall into four main families. *Contrastive alignment* uses dual encoders to learn a shared embedding space with InfoNCE-style losses, as in ALIGN and CLIP [8], [25]. *Fusion objectives* rely on bi- or cross-encoders, where ITM and MLM losses support fine-grained

TABLE 7. Summary of evaluation metrics for visual language models.

Metric	Target Task	Strengths	Weaknesses
BLEU [28]	Captioning, text generation	Simple, widely used, measures n -gram precision	Insensitive to synonyms, poor correlation with human judgment
METEOR [29]	Captioning, summarization	Accounts for synonyms, stemming, word order; better alignment with human evaluation	Still surface-form oriented, less robust for long texts
ROUGE [30]	Summarization, captioning	Recall-focused; good for coverage-oriented tasks	Not semantically aware, weak for multi-modal reasoning
CIDEr [31]	Image captioning	Consensus-based; designed for captioning; strong correlation with human scores	Requires multiple references; less generalizable beyond captioning
Recall@k [32]	Cross-modal retrieval	Directly measures retrieval quality; intuitive and interpretable	Does not capture semantic relevance beyond ranking position
MRR (Mean Reciprocal Rank) [33]	Retrieval, matching	Considers rank position; useful for ranked retrieval tasks	Less informative for large candidate sets compared to Recall@k
nDCG [34]	Retrieval with graded relevance	Accounts for varying degrees of relevance; robust for ranked outputs	Requires graded annotations; harder to collect
VQA Accuracy [22]	Visual Question Answering	Task-specific; measures factual correctness of answers	Binary scoring; ignores partial correctness or reasoning quality
Exact Match / F1 [43]	QA, instruction following	Clear, objective measure for textual answers; F1 captures partial matches	Insensitive to paraphrases; limited for open-ended generation
Human Judgments [44]	Any generation or reasoning task	Captures fluency, coherence, usefulness and factuality; gold standard for quality	Expensive, time-consuming, subjective; low reproducibility
Pairwise Preference Studies [44]	Comparative evaluation	Provides relative model ranking; reduces subjectivity by direct comparison	Requires large-scale human studies; not scalable for many models

alignment (LXMERT, ViLT) [6], [17]. *Generative objectives* adopt encoder–decoder setups, using next-token likelihood for captioning and QA or instruction tuning (Vid2Seq, MiniGPT-4) [10], [19]. Finally, *multitask hybrids* combine contrastive, ITM/MLM and generative losses, as in Flamingo and BLIP [7], [18].

In practice, contrastive training scales well to retrieval and zero-shot transfer, fusion supports token-level reasoning, generative or instruction tuning enables open-ended capability and hybrids trade training complexity for broader coverage.

C. MODALITY INTERACTION STRATEGY

Early fusion concatenates visual and text tokens at the input (simple, less scalable). Late fusion combines unimodal decisions (efficient, weak grounding). Cross-attention fusion iteratively conditions each modality on the other (e.g., LXMERT, Flamingo) [6], [18], delivering stronger semantic alignment. Retrieval-only alignment (Two-Tower) aligns pooled embeddings for search/zero-shot without token-level fusion [8], [25].

D. INPUT GRANULARITY & PERCEPTUAL FRONT-ENDS

VLMs differ in the granularity of their perceptual front-ends. At the *global* level, image-level features enable efficient zero-shot classification, as demonstrated in CLIP [5], [25]. *Region-level* representations use detector-driven object features to support object-centric reasoning, as in LXMERT. *Patch-level* encodings rely on ViT-style patches, which balance fine-grained reasoning with scalability (e.g., BLIP, Flamingo) [7], [18]. Finally, *OCR/text tokens* capture explicit

text-in-image signals, which are critical for tasks such as TextVQA [23].

E. ROBUSTNESS & BIAS PROPERTIES BY DESIGN

Architectural choices interact with robustness and bias (see Table 8). For example, Arias et al. [51] reveal CLIP’s bias towards gray colors and textual data over data images, highlighting the requirement to closely examine color and other perceptual properties in embed-heavy designs. Two-Leg/One-Tower merge can potentially assist grounding but may increase compute and sensitivity to domain shift; LLM-grounded stacks maintain language model priors, so their safety in instruction and avoidance of hallucination are keystone factors

In the following sections we illuminate the two axes most prominently guiding system behavior and comparability—*architectures* and *training objectives*—and save data regimes, metrics and applications until Sections V–VII. This independence enables us to examine design trade-offs without having to resort to benchmark churn or domain-specific constraints.

IV. MODEL ARCHITECTURES AND ADVANCES

This section elaborates on developments *outside the core families* from Section III, grouped based on the system property addressed primarily by each development: efficiency, adaptive behavior with restricted monitoring, interpretability, on-device inference, knowledge augmentation and embodied planning. Table 11 displays these developments and assigns benefits to previously-introduced KPIs from Tables 7 and 8.

TABLE 8. Dimensions of robustness evaluation beyond academic benchmarks for Visual Language Models.

Dimension (Representative refs)	Description	Challenges for VLMs
Domain Shift (e.g., <i>medical adaptation</i> [45], <i>RS CLIP-MoA</i> [26], <i>MapReader</i> [46])	Evaluating models on data from unseen domains (e.g., medical images, satellite imagery, cartography, social media)	Models often overfit to benchmark distributions and degrade sharply on out-of-distribution inputs; requires domain adaptation and task-specific fine-tuning
Adversarial Robustness (e.g., <i>VL-Trojan backdoors</i> [15])	Resistance to imperceptible perturbations or maliciously crafted examples	Dual-encoders (e.g., CLIP-like) and multimodal LLMs are vulnerable to small adversarial changes or backdoor triggers in images/prompts, leading to misclassification or hallucination
Bias and Fairness (e.g., <i>dataset audits</i> [47])	Testing for demographic, cultural or linguistic biases	Benchmarks rarely reflect global diversity; VLMs risk amplifying social or cultural stereotypes inherited from web-scale data
Noisy / Incomplete Inputs (e.g., <i>web-scale noise</i> [20], <i>modality guidance for robustness</i> [14])	Handling corrupted, missing or ambiguous multimodal inputs	Models often rely on superficial correlations; performance drops when visual/textual inputs are partially degraded, misaligned or noisy
Interpretability and Transparency (e.g., <i>multimodal CoT</i> [2], <i>expert-based evaluation in CCBench</i> [48])	Ability to explain predictions and reasoning	Current VLMs provide fluent outputs but limited visibility into decision-making; lack of faithful rationales hinders trust in safety-critical settings
Resource Efficiency (e.g., <i>lightweight ViLT</i> [17], <i>efficient hybrid mixing</i> [49], <i>real-time FPS in apps</i> [50])	Performance under constrained computational or memory budgets	Academic benchmarks assume abundant resources; deployment on edge/mobile requires lightweight, efficient, yet robust models without sacrificing accuracy

A. RECENT ADVANCES BY SYSTEM PROPERTY

Quadratic attention places constraints on deployability and context. VisualRWKV-HM [49] substitutes full self-attention with recurrent/state and channel mixing, with near-SOTA results but much lesser memory and latency. In our taxonomy it lies in the “Unified/Hybrid” family (Table 10); significant KPIs include increased throughput and lesser memory with comparable accuracy, with an energetically advantaged profile.

Prompt-based adaptation is a light lever for domain/task shift. CuTCP [52] uses *LLM-generated, class-aware* prompts, improving known/unknown discrimination and fine-grained categories. BIP [53] introduces *bi-modality individual-aware* prompts via textual/visual knowledge embeddings, coupling class-aware text with instance-aware visual prompts across four tasks and 15 benchmarks. These complement Two-Tower/Two-Leg families as parameter-efficient layers; KPIs: accuracy/F1 gains with negligible parameter growth and improved OOD behavior.

Under partial labels, naive CLIP-style training entangles semantics. Ruan et al. [54] address multi-label recognition with partial labels via a semantic decoupling module (category-specific spatial attention) and category-specific prompt optimization, improving mean AP/F1 and label-wise calibration with lighter models; this strengthens Two-Tower/Two-Leg training under weak supervision.

For high-stakes application, post-hoc saliency is also inadequate. Fang et al. [55] introduce a decomposed concept bottleneck model (DCBM) with *learned visual concepts* in place of predefined text concepts while maintaining competitive precision but with enhanced text–image groundability (“Hybrid/Emerging” with an explicit interface to

interpretability). KPI deltas are concept fidelity, grounding agreement and human preference.

EdgeVL [58] combines knowledge distillation with dual modality and contrastive learning with quantization awareness to compress VL models without human annotation to an edge device-applicable level while preserving the quality of quantized features. Weng et al. [59] embed bilingual features in a shared space to boost captioning, MT and VQA with low-resource multilingual data. This is an orthogonal deployment layer.

Hallucination is alleviated through retrieval and long-tail knowledge is augmented. RAVL [56] simultaneously adapts retriever and VLM for VQA with knowledge-based retrieval of (image, question, evidence) tuples, having SOTA on OK-VQA. Acting as an objective layer over Two-Leg/Encoder–Decoder stacks, it contributes to VQA quality and attribution quality improvement and robustness augmentation towards knowledge gaps.

In addition to perception, there is planning and acting necessary for agents. Multi-agent embodied planning with commonsense guidance is suggested by Brienza et al. [57] in single-view worlds, with PG2S evaluator for plan quality introduced (tested on ALFRED). As an application-level composition of VLM perception and LLM reasoning, KPIs include plan success/quality (PG2S), sample efficiency and robustness to ambiguity in the scene.

Techniques have to be chosen by deployment constraints or data conditions and report the KPIs they primarily affect (Table 11). Linear/hybrid mixing serves tight latency/memory budgets; prompt-based PEFT and semantic decoupling fit scarce/partial labels; concept bottlenecks support explainability; distillation/quantization and cross-lingual projection enable edge/multilingual scenarios; retrieval

TABLE 9. Comparison of vision–language survey articles and our contribution.

Survey	Primary Scope / Angle	Modalities	Tasks Coverage	Model Era & Objectives	Datasets & Eval	Ethics Bias	What's Missing (covered by us)
Baltrušaitis et al. [4]	Foundational taxonomy of <i>multimodal ML</i> (representation, translation, alignment, fusion)	Vision+Text (+others)	Broad, classic (captioning, VQA, retrieval)	Pre–foundation era; shallow/deep fusion (little on contrastive/generative pretraining)	Limited benchmark synthesis	Limited	Foundation-model era, instruction tuning, unified VLMs, domain apps
Du et al. [35]	<i>Vision–language pre-trained models</i> (VLP) focus	Vision+Text	Captioning, VQA, retrieval (pretraining-centric)	Early CLIP/ALIGN epoch; contrastive/ITM/MLM emphasized	Partial	Limited	Post-2022 LLM-decoder VLMs, instruction-following, app-level KPIs
Li et al. [36]	Broad <i>vision–language intelligence</i> : tasks, repr. learning, large models	Vision+Text	Wide task taxonomy	Early foundation trends; objectives discussed	Partial overview	Limited	Deeper evaluation unification, datasets→objectives linkage, robustness
Xu et al. [38]	<i>Multimodal Transformers</i> (architectures)	Vision+Text (+speech)	Broad, model-architecture centric	Transformer variants; pretraining objectives surveyed	Partial	Limited	Application grounding, KPI frameworks, domain shifts, security
Chen et al. [37]	<i>VLP</i> : methods and pretraining strategies	Vision+Text	Captioning, VQA, retrieval (VLP-focused)	Contrastive / ITM / MLM; some generative	Partial, method oriented	Brief	Cross-walking to apps/metrics; unified tables for data synthesis
Zhang et al. [5]	VLMs for <i>vision-centric tasks</i>	Vision+Text	Detection /segmentation /vision tasks	Applying VLMs to vision tasks; objectives by use	Strong on vision benchmarks	Limited	Broader application modalities (dialogue, reasoning), dataset KPIs, applications mapping
Our survey	Unified taxonomy + practical KPIs bridging architectures and objectives to datasets, metrics and applications	Vision+Text	End-to-end: captioning, VQA, retrieval, dialogue, video, domain apps	Full spectrum: contrastive, ITM/MLM, generative, instruction tuning; unified/generalist VLMs	Analytical: dataset KPIs; eval metrics table; supervision → objective mapping	Bias/robustness/security dimensions (Table 8)	New comparative tables (architectures, timelines, data synthesis), cross-links across Sec. V–VII

augmentation addresses knowledge gaps; and planner-aware compositions target embodied tasks.

Rule-of-thumb bullets:

- *Need speed/footprint?* Prefer linear/hybrid mixing; keep fusion shallow; measure latency, memory and energy.
- *Few labels or shifting domains?* Start with PEFT prompts (class-/instance-aware) or semantic decoupling; track OOD gaps and calibration.
- *Must justify outputs?* Insert a concept bottleneck; report grounding/attribution alongside accuracy.
- *Edge or multilingual?* Distill and quantize; add cross-lingual alignment; report FPS and multilingual parity.
- *Missing knowledge?* Add retrieval and score evidence; prefer selective abstention over confident guesses.
- *Embodied tasks?* Compose VLM perception with a planner and evaluate with plan-quality metrics, not just task accuracy.

V. TRAINING TECHNIQUES AND DATASETS

VLM’s performance is conditioned on *how* we tune (objectives, adapters, prompting, robustness) and *what* we learn from (curated, web-scale, bootstrapped or domain sets). This section organizes the space as (i) training methodologies (objective-/parameter-/compute-/data-oriented) and (ii) dataset regimes and generation, with explicit references to Section II and Table 12.

A. TASK-LEVEL OBJECTIVES AND INSTRUCTION TUNING

As summarized in Section III-B, objective choice follows the architecture:

1) OBJECTIVE VARIANTS FOR BETTER ALIGNMENT

VTPL [60] jointly learns *visual* and *text* prompts and augments contrastive learning with Poly-1 InfoNCE plus a center loss, improving few/zero-shot transfer across 11 image

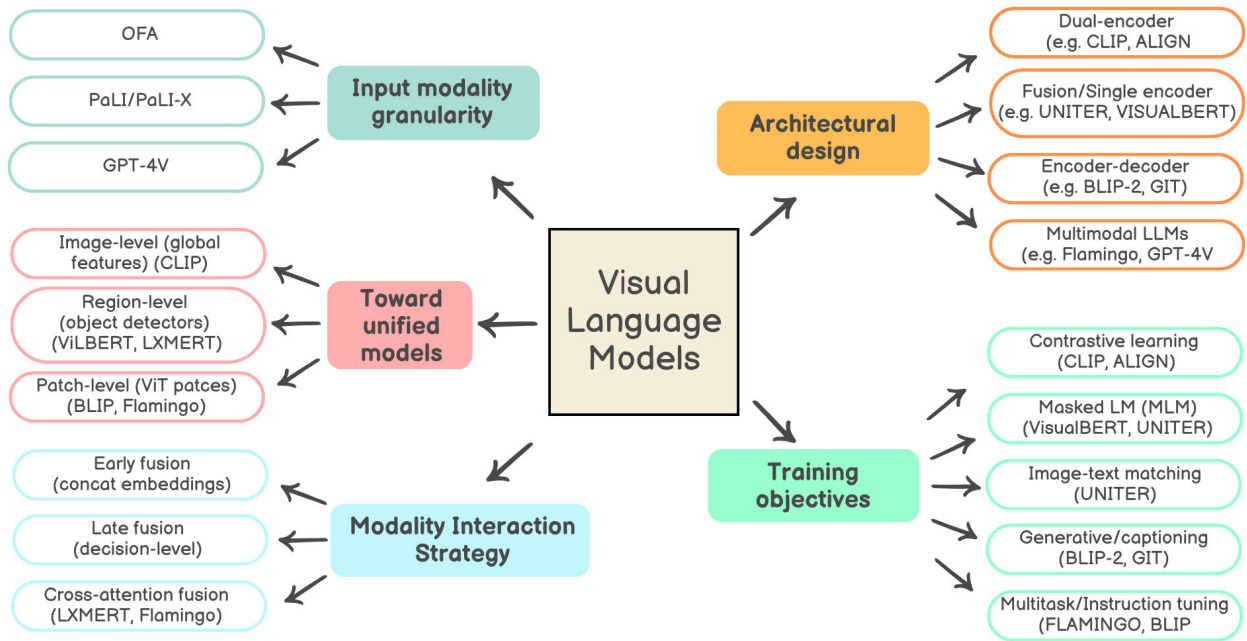


FIGURE 2. Taxonomy of Visual-Language Models (VLMs). The branches follow architectural family (Two-Tower / Two-Leg / One-Tower / Encoder-Decoder), training objectives, interaction strategy and input granularity.

datasets. DoCo [61] adds document-object contrastive signals for text-rich imagery, strengthening fine-grained grounding without inference overhead.

Takeaway: maintain contrastive for web-scale pairs but stabilize with class/instance regularizers and prompt learning.

2) INSTRUCTION TUNING (MAKING MODELS FOLLOW TASKS)

Instruction layers are typically added to generative stacks: Flamingo and MiniGPT-4 [10], [18]; domain variants such as Diagnostics-LLaVA and DDC-Chat [27], [68] inherit the same recipe. This enables multi-task prompting, few-shot adaptation and interactive use.

Takeaway: pair AR decoding with corpora of instructions; report exact-match/F1 and human preference with capability metrics.

3) SEMI/WEAK SUPERVISION (WHEN LABELS ARE SCARCE)

SemiCD-VL [69] uses VLMs to pseudo-label change detection; annotation-free detection [70] replaces manual labels with VLM-generated signals to kick-start training.

Takeaway: use model-in-the-loop labels to expand coverage, then refine with small curated sets for calibration.

4) RULE OF THUMB

Web-scale noisy pairs $\Rightarrow$ *contrastive*; curated or instruction-rich data $\Rightarrow$ *ITM/MLM* + *AR*; mixed/bootstrapped sources $\Rightarrow$ *hybrid multitask*, optionally with prompt/adapters for stability.

B. OPTIMIZATION & PARAMETER-EFFICIENT ADAPTATION

We group techniques by *what they optimize for*: (i) parameter-efficient prompting/adapters, (ii) learning under weak/partial labels, (iii) robustness to shift/adversaries and (iv) training curricula & modular bridges. Table 13 summarizes the recipe, when to use it and KPIs to report.

C. COMPUTE- AND DEPLOYMENT-AWARE TRAINING

Goal Make VLMs practical under *rate/latency/memory* constraints and heterogeneous devices (CPU/GPU/edge) while retaining grounding quality.

One can compress and regulate bitrates with token-aware compression on big VLMs [65], which jointly co-trains a pre-editor and a learned codec to swap bits with fidelity while obeying some deployment budgets and varying bitrates. At reporting time, report your end-to-end latency, memory usage, energy per sample and your task-score vs. bitrate curves.

Edge and multilingual inference are aided by Edge VL [58], distilling contrastive learning with quantization awareness to retain quality of features during quantization on edge devices. Lightweight, multilingual VLMs [59] further embed non-English and English features in a common space, enhancing caption, MT and VQA in low-resource settings. Reports must include model size, FPS/latency, multilingual parity (per-language).

As a rule of thumb, compress *tokens* rather than only weights, pair QAT with contrastive awareness and always publish latency and energy alongside accuracy.

TABLE 10. VLM architectural families with objectives, supervision, tasks, pros/cons and representative models.

Family	Architectural Sketch	Primary Objectives	Typical Supervision	Canonical Tasks (Pros / Cons)	Representative Models
Two-Tower	Separate image/text encoders; pooled embeddings aligned	Contrastive (InfoNCE)	Web-scale alt-text (weak)	Retrieval, zero-shot (scales well / limited token-level grounding)	CLIP [25]; ALIGN [8]
Two-Leg	Dual streams with cross-attention	ITM/MLM; optional AR	Curated captions/Q&A	VQA, grounding (fine-grained / heavier compute)	LXMERT [6]; BLIP (fusion) [7]; Flamingo (interleaved) [18]
One-Tower	Single multimodal transformer (patch + word tokens)	ITM/MLM; light AR	Curated/cleaned captions	Captioning, QA (simple/efficient / may lag on fine-grained)	ViLT [17]
Encoder–Decoder	Vision encoder → decoder-only LLM	AR (caption/QA); instruction tuning	Mixed + instruction data	Open-ended dialogue (flexible / hallucination risk)	MiniGPT-4 [10]; Vid2Seq [19]
Unified / Hybrid	Mix contrastive ITM/MLM + AR	+ Mixed (bootstrapped + curated)	Retrieval + captioning + VQA in one stack (versatile / complex optimization)	BLIP/BLIP-2 [7], [42]	

TABLE 11. Advances beyond core families: what they claim, how they work, where they fit and practical trade-offs.

Theme	Representative Method	Core Idea	Primary Gains (KPIs)	Trade-offs / Risks	Refs
Efficiency (Linear/Hybrid)	VisualRWKV-HM	Linear hybrid mixing (state/channel) vs. quadratic attention	Throughput↑, memory↓, energy↓	New training dynamics; maturity of tooling	[49]
Prompting & Personalization	CuTCP; BIP	LLM-generated class-aware prompts; bimodality instance/class prompts	Acc/F1 on FGVC, OOD generalization	Prompt brittleness; fairness drift	[52], [53]
Partial/Weak Supervision	MLR-PL (decouple+cat-prompt)	Category-specific spatial decoupling + prompt alignment	mAP/F1↑, calibration	Extra modules; tuning complexity	[54]
Interpretability	DCBM	Learned visual concepts as bottlenecks	Groundability, concept fidelity, human pref.	Capacity loss if overly constrained	[55]
Edge & Multilingual	EdgeVL; Multilingual light VLM	Distillation + QAT; cross-lingual shared space	Size/latency↓, multilingual acc.↑	Perf. loss at extreme compression; lexicon bias	[56], [57]
Retrieval-Augmented	RAVL	Jointly tune VLM and retriever with evidence	OK-VQA acc.↑, attribution	Retriever drift; latency	[58]
Embodied Agents	Multi-agent + PG2S	Commonsense-guided planners; plan-quality metric	Plan quality/success; sample efficiency	Sim2real gap; evaluation variance	[59]

D. KNOWLEDGE AND RETRIEVAL AUGMENTATION

Retriever + generator, trained together. RAVL [56] couples a retriever with a VLM for knowledge-based VQA, fine-tuning both on *evidence tuples* and adapting the retriever via VLM signals. This yields +3.63% on OK-VQA over prior SOTA and improves attribution. *Report:* answer accuracy, evidence precision/recall and ablation of joint vs. frozen retriever.

Practice. Use RAG when answers require world knowledge; include *selective abstention* and show coverage–accuracy curves.

E. DOMAIN-AWARE TRAINING RECIPES

In ecologic monitoring, Eco-VLMs [66] present species-conscious text augmentation alongside a random contrastive loss, achieving F1 improvements of 3.8–4.4 points. When presenting results in this scenario, report per-species F1 and out-of-d.

Few-shot methods like FTPA + DMM [67] adapt text prompts and dynamically adjust features and reach state-of-the-art AUROC on MVTec-AD and VisA. The reports must include AUROC/AUPRC and few-shot budget with the results of line/lot shift tests.

TABLE 12. Training techniques at a glance: what they change, when to use them, KPIs impacted and exemplar references.

Technique Family	What it Changes	When to Use	KPIs Impacted	Exemplars
Objective variants (contrastive+)	Losses (Poly-1 InfoNCE, centers)	Weak/noisy pairs; better alignment	Zero-shot, Recall@k, calibration	VTPL [60], DoCo [61]
Instruction tuning	Decoder rules; task formats	Open-ended IF; multi-task prompts	VQA acc., IF success, human pref.	Flamingo/MiniGPT-4 [10], [18]
PEFT via prompting	Small prompt params	Few/zero-shot; OOD; class granularity	F1/mAP; robustness	TCP/CuTCP/BIP [52], [53], [62]
Partial/weak supervision	Feature decouple + cat-prompts	Partial labels; multi-label	mAP/F1; calibration	MLR-PL [54]
Robustness finetuning	Vis-guided prompts; robust aug.	Adv. robustness; dist. shift	Acc. under attack; POPE/CHAIR	MDAPT [63], Robust-MixGen [64]
Retrieval augmentation	External knowledge pathway	KB-VQA; long-tail facts	VQA acc., attribution	RAVL [58]
Compression/distillation	Token rate; quantization	Edge/latency/size budgets	Latency, memory, energy	Var. bitrate [65], EdgeVL [56]
Domain-aware recipes	Data aug.; loss tweaks	Ecology, industrial FSAD	F1/AUROC; generalization	Eco-VLM [66], FTPA+DMM [67]

TABLE 13. Scenario → recommended recipe, primary KPIs and exemplars.

Scenario	Recommended Recipe	KPIs to Watch	Refs
Throughput / low memory	Linear or hybrid mixing; slim fusion	Latency, memory, energy	VisualRWKV-HM
Labels scarce / domain shift	Prompt-based PEFT; class/instance-aware prompts; semantic decoupling	F1/mAP, calibration, OOD gap	CuTCP, BIP, MLR-PL
Explainable decisions	Concept bottlenecks; grounded heads	Groundability, concept fidelity, human pref.	DCBM
On-device / multilingual	Distillation + QAT; cross-lingual projection	Size, latency/FPS; multi-lang acc.	EdgeVL; light multilingual VLM
Knowledge heavy / long tail	Retrieval-augmented generation with evidence scoring	VQA acc., attribution precision/recall	RAVL
Acting in the world	Perception + planner; plan-quality evaluation	Task success, PG2S, sample efficiency	PG2S (multi-agent)

To mitigate hallucination and bias, RBD [71] re-balances decoding to curb text dominance via controlled noise and visual salience, improving CHAIR/POPE; dataset audits and color-bias analyses [47], [51] further reveal modality skew. Here, include CHAIR/POPE, subgroup gaps and qualitative examples with visual attribution. In general, it is recommended to start with adapters or domain prompts, assess domain shift overtly and put capability metrics alongside reliability add-ons (Sec. VI).

F. DATASETS: REGIMES, CONSTRUCTION AND KPIS

Table 5 groups supervision sources into curated, web-scale, bootstrapped, instruction and domain-specific categories. Web-scale pairs are best paired with *contrastive* training; curated or instruction-rich data align with *ITM/MLM plus generative* objectives; hybrid or bootstrapped sources motivate *multitask* setups.

Representative sets span multiple regimes. Curated resources include COCO, VQA v2 and TextVQA [21], [22], [23]. Web-scale collections include Conceptual Captions and LAION-5B [20], [24]. Bootstrapped corpora arise from BLIP-style filtering and rewriting [7]. Specialized benchmarks cover OK-VQA, ChipVQA and MapTrain [9], [16], [46]. Video and narration-driven supervision is exemplified by Vid2Seq pretraining [19]. Application-oriented sets include IMFND for misinformation and Diagnostics-LLaVA or DDC-Chat for automotive diagnostics and driving behavior [14], [27], [68].

Important dataset KPIs (Table 14) are scale, diversity of domain and language, quality of annotation, coverage of tasks, bias and fairness [47] and reproducibility or accessibility. COCO has high-quality labels with modest scale and diversity; LAION-5B has immense scale and diversity but is noisier and biasser; specialized collections like ChipVQA

and the visualization-literacy suite [72]. Construction trends range from human-curated instructions and annotations to web scraping with language or filters (NSFW), model-in-the-loop bootstrapping as in BLIP, synthetic or augmented evidence as in retrieval-augmented pipelines, to domain-expert authoring such as in medicine or remote-sensing.

G. CHALLENGES & MITIGATIONS IN DATA AND TRAINING

This noisy and bias-rich web data can be addressed with filtering or model-in-the-loop bootstrapping or dataset audits [47]. Augmentation methods like RobustMixGen are also more robust [64].

Balancing modal imbalance and related hallucinations are best tackled with re-balancing decoding (RBD) [71] and with retrieval augmentation (RAVL) integrated in such a manner that responses are external evidence-based [56]. In order to combat domain shift, especially in remote sensing and medicine, practitioners generally deploy adapters and mixed recipes of training (e.g., CLIP-MoA or multi-level adapters) [26], [45].

Compute and latency constraints can be alleviated with token-aware compression [65] and with distillation or quantization for edge deployment [58]. Finally, when labels are scarce, prompt-based parameter-efficient tuning (TCP, CuTCP, BIP) [52], [53], [62] and semi-/annotation-free pipelines [69], [70] are effective low-cost alternatives.

VI. EVALUATION AND BENCHMARKING

Judging VLMs needs to find a balance between *capability* (can the model do the task?) and *reliability* (does it do the task consistently, accurately, and for the right reasons?). We organize this section along these two axes and standardize protocols, measures and reporting so results are reproducible and comparable. We refer to metric families in Table 7 and robustness dimensions in Table 8.

As shown in Tables 12, 15, and 16, architectural design, training knobs, and optimization strategies collectively determine VLM reliability and generalization. Comprehensive benchmarking should therefore report not only performance but also contextual metadata such as dataset provenance, parameter count, energy consumption, and robustness drop under adversarial or compositional perturbations. By standardizing these reporting practices, future benchmarks can foster reproducible and interpretable progress across the VLM research landscape.

A. WHAT TO MEASURE (CAPABILITY VS. RELIABILITY)

We assess *capability* through captioning quality—using BLEU, METEOR, ROUGE and CIDEr [28], [29], [30], [31]—as well as retrieval effectiveness via Recall@k, MRR and nDCG [32], [33], [34]. We additionally report VQA accuracy [22] and, for instruction following or dialogue, exact match or F1 together with human preference judgments.

While ROUGE/CIDEr are widely adopted for captioning, they are primarily surface-overlap measures and may not adequately reflect *semantic understanding* or *cross-modal*

logical reasoning. For reasoning-heavy settings, we recommend complementing them with task-specific correctness checks (e.g., answer accuracy/consistency), semantic metrics and structured evaluations (step-wise rationales, grounding/evidence attribution), alongside targeted human or expert review.

Reliability is captured by performance under distribution shift and noise, adversarial robustness, fairness and bias analyses, calibration and abstention behavior, hallucination and factuality checks and efficiency considerations (latency, memory, energy). Definitions and representative threats appear in Table 8.

B. CANONICAL BENCHMARKS AND PROTOCOLS

For captioning, we rely on MS COCO and related datasets and report BLEU, METEOR, ROUGE and CIDEr (see Section “Datasets: Regimes, Construction and KPIs”). Retrieval is typically evaluated on Flickr30k and MS COCO using Recall@k, MRR and nDCG. VQA performance is measured on VQA v2, TextVQA and OK-VQA, with accuracy as the primary metric; OK-VQA additionally probes external knowledge [9], [22], [23]. Instruction following is evaluated with MME, MMBench, MMMU and application-specific suites with exact match or F1 and human preference scores as outputs.

When describing protocols, it is important to indicate whether the setup is zero-shot, few-shot (with k and prompt information) or fine-tuned. Other parameters to be defined include in-context exemplars. OCR enabling and image resolution, decoding parameters (beam size and temperature) and development and testing splits. Data de-duplication with respect to pretrained corpora should also be explained to avoid leakage.

C. BEYOND LEADERBOARDS: DIAGNOSTIC & STRESS TESTING

Security and jailbreak resistance remain open concerns: Samuel et al. [75] show that multimodal prompts can bypass guardrails, while DiffPure pre-processing improves resilience. Visualization literacy is another weakness: Pandey et al. [72] find that VLMs struggle with anomalies and misleading charts despite strong performance on routine reading tasks. Cognitive probes further reveal deficits in relations, composition and contextual reasoning; light tuning helps alleviate these gaps [76]. Finally, Ma et al. [77] introduce ZSIR to evaluate zero-shot cognition alignment by measuring human-like similarity without finetuning.

D. RELIABILITY BATTERIES: ROBUSTNESS, SAFETY AND CALIBRATION

Adversarial and noise robustness should also be quantified by reporting accuracy on standard corruptions and, when relevant, by reporting adversarial budgets and Expectation-Over-Transformation settings; summarize degrade over clean inputs. Hallucination and factuality should be monitorable

TABLE 14. Unified overview of datasets and their use across domains (links verified Oct. 1, 2025).

Domain	Representative Datasets (official pages)	Capability Metrics	Reliability Add-ons	Typical Patterns
General Benchmarks (captioning/VQA/pretraining)	COCO ; VQAv2 ; Conceptual Captions ; OK-VQA ; TextVQA ; LAION-5B	BLEU, CIDEr, Acc/F1; Recall@k	Bias audits; contamination checks	Encoder–Decoder; Two-Tower; retrieval-augmented pretraining
Meta-analysis & Audits	Birhane et al. (Dataset Audit)	Bias prevalence; stereotype frequency	Provenance tracking; licensing audits	Dataset governance; documentation practices
Medical & Pathology	VQA-RAD ; PathVQA ; ROCO	AUC/Accuracy; VQA accuracy; Recall@k	Expert agreement (ECE); abstention; shift tests	Two-Leg/Encoder–Decoder; adapters; instruction tuning
Information Integrity & Security	NewsCLIPPings ; Fakeddit	Acc/F1 (FND); ASR (attacks)	Robustness to multimodal attacks; attribution/traceability	Two-Leg with instruction tuning; retrieval-augmented variants
Transportation & Mobility	100-Driver DDC	Acc/Recall/F1; latency/FPS	Stress under blur/low-light; on-device constraints	Encoder–Decoder (LLM backbone); skill-activated reasoning; edge offloading
Agriculture & Environment	Fruits-360 ; PlantVillage ; iNaturalist	mAP@0.5; F1; FPS	Weather/illumination robustness; OOD varieties	Two-Tower + prompt tuning; domain augmentations
Video Understanding & Surveillance	Kinetics-700 ; MSR-VTT	Top-1/Top-5; Recall@k; BLEU/CIDEr	Temporal corruption tests; privacy; false alarm cost	Encoder–Decoder with visual prompting; interleaved cross-attn; multi-task finetune
Manufacturing & Industrial QA	MVTec AD ; KolektorSDD2 ; Magnetic Tile Defects	AUROC/AUPRC; task success; retrieval	Shift across lines/lots; explainability of defects	Two-Leg/Unified; adapters; domain-specific prompting
Remote Sensing & Cartography	BigEarthNet ; SpaceNet ; xView	Accuracy; zero-shot transfer; IoU	Cross-sensor/region shift; geo-bias audits	Two-Leg with adapters; contrastive pretrain + instruction adaptation
Multimodal Agents (GUI/Docs)	Mind2Web ; ChartQA ; DocVQA	Task success; steps-to-goal; VQA accuracy	Usability; refusal/abstention; latency constraints	Encoder–Decoder; high-res encoders; OCR-aware inputs
Cross-domain / Synthetic	Visualization Benchmark ; Literacy (VisLit) ; RefChartQA ; AI2D	Accuracy on vis. literacy; BLEU/CIDEr; instruction-following success	Human audits; synthetic→real robustness	Instruction-tuned Encoder–Decoder; adapters; synthetic pretrain + real finetune

with CHAIR/POPE on captioning and Q&A and on knowledge-based VQA attribution—evidence precision and recall—should be quantified especially with retrieval augmentation like RAVL [56]. Bias and fairness analysis should also adhere to Table 7, investigating demographic, cultural and linguistic skew and reporting subgroup metrics and gaps. Calibration and abstention require explicit reporting: report ECE or Brier score, include coverage–accuracy curves on selective prediction and permit “I don’t know” responses so safe abstention may be estimated. Finally, efficiency should be measured with end-to-end latency, memory footprint and, when feasible, energy per sample, all related with intended deployment constraints.

E. HUMAN EVALUATION AND EXPERT STUDIES

Pairwise preferences and rubric-based scores in terms of fluency, helpfulness, factuality and safety should be utilised. For domain tasks (medical, industrial, RS), *expert* annotators should be recruited and inter-annotator agreement (Cohen’s κ / Krippendorff’s α), adjudication policy and cost/time should

be reported. Single-rater studies in safety-critical should be avoided.

F. REPORTING STANDARDS AND REPRODUCIBILITY

Prompts, decoding settings, seeds, the resolution of images, OCR use and all corpora retrieval should always be released. Confidence intervals (bootstrap or approximate randomization), carry out significance tests from head-to-head comparisons and reveal compute budget (train/infer FLOPs), model size and pretraining data provenance/filters (data cards) also have to be reported. Finally, checks for document contamination over pretraining and evaluation splits should be done.

Capability and reliability should be connected with Section VII: For instance, medical/pathology $\Rightarrow$ expert accuracy, calibration, abstention; industrial inspection $\Rightarrow$ AUROC under shift; latency/edge budgets; remote sensing $\Rightarrow$ domain shift and evidence attribution; misinformation $\Rightarrow$ source grounding, error severity. In cases where possible, include *task-focusing* KPIs as well as generic ones.

TABLE 15. Summary of visual language model architectures and advances.

Model / Framework	Type	Main Contribution	Key Insight / Result
VisualRWKV-HM [49]	Linear VLM	Hybrid mixing mechanism with linear complexity	State-of-the-art performance, high efficiency, scalable encoder/decoder
CuTCP [52]	Prompt Tuning VLM	Class-aware prompt generation via LLMs	Better generalization and fine-grained classification than TCP
CLIP-based MLR [54]	Dual-Encoder VLM	Semantic decoupling and prompt optimization for partial labels	Outperforms CLIP on multi-label recognition with partial labels
DCBM [55]	CBM + VLM	Decomposed visual concepts for interpretable classification	Matches closed-box performance with interpretable outputs
EdgeVL [56]	Lightweight VLM	Distillation + quantization for edge deployment	Reduced model size, improved open-vocabulary classification on devices
LGVLM [57]	Cross-lingual Gen. VLM	Lightweight multilingual image-text model	Better multilingual performance, efficient captioning and VQA
RAVL [58]	Retrieval-Augmented VLM	Integrates retrieved knowledge in VQA	Improves OK-VQA accuracy by 3.63%, alleviates visual info loss
BIP [53]	Prompt Tuning VLM	Instance- and class-aware prompt embeddings (TKE/VKE)	Outperforms 15 benchmarks across 4 tasks, strong generalization
Multi-agent VLM [59]	Embodied Planning	Multi-agent planning via commonsense + image input	Introduces PG2S metric, improved task planning in ALFRED dataset
CLIP [25]	Dual-Encoder	Contrastive learning with ViT and text encoder	Efficient retrieval, zero-shot generalization
ALIGN [8]	Dual-Encoder	Large-scale noisy data for pretraining	Strong generalization despite weak supervision
UNITER / VisualBERT	Fusion Model	Unified visual-linguistic pretraining	Effective for VQA and alignment tasks with object detector input
ViLT [17]	Fusion Model	Detector-free, patch-based input	Reduces compute, competitive on VQA and retrieval
BLIP-2 [42]	Encoder-Decoder	Modular vision encoder + Q-former + LLM	Efficient image-text generation with frozen components
GIT [73]	Encoder-Decoder	Unified end-to-end seq2seq training	Strong performance on captioning, storytelling
Flamingo [18]	Encoder-Decoder	Few-shot multimodal generation via frozen LLM + adapters	Minimal supervision, strong few-shot VQA and dialogue
PaLI / PaLI-X [74]	MLLM	Unified multilingual-multimodal seq2seq	Versatile on captioning, OCR, VQA tasks
MiniGPT-4 [10]	MLLM	Lightweight alignment of vision encoder to LLM	General-purpose reasoning with low resource use
GPT-4V	MLLM	Native multimodal input in GPT-4	Grounded generation, diagram understanding, general-purpose vision-LM

TABLE 16. Training/deployment knobs: what they change, when to use and what to report.

Dimension	Options / Methods	When to Use & KPIs
Objectives	Contrastive [8], [25]; ITM/MLM [6], [17]; AR [10], [19]; hybrids [7], [18]	Match supervision; report Recall@k, VQA acc., BLEU/CIDEr
PEFT (prompt/adapters)	Class/instance prompts [52], [53]; Q-Former [42]; LoRA	Few labels / OOD; F1/mAP; calibration
Robustness	Robust aug.; adversarial-aware finetune	Stress tests; ASR; robustness drop
Compression	Distillation; QAT [56]; token-rate control [65]	Edge/latency; size, FPS, energy/sample
Retrieval	Joint retriever-generator [58]	Knowledge tasks; evidence precision/recall

Leaderboard scores are necessary but insufficient. Performance indicators (Table 7) should be combined with reliability tests or measures (Table 8) to evaluate both capability and dependability. Standardized protocols (Table 17) should be followed and the reporting checklist should be used to ensure comparability and trustworthy claims.

VII. APPLICATIONS

VLM applications span safety-critical domains (medicine, industrial QA), open-world perception (remote sensing, video), information integrity (mis/disinformation) and human-computer interaction (multimodal agents) (Table 18). To make this section actionable, we (i) define an applications

taxonomy, (ii) map each domain to *datasets*, *capability metrics and reliability add-ons* (Table 14) and (iii) summarize *system patterns*—typical architecture/objective choices and deployment constraints. This cross-walk aligns with *super-vision regimes* (Table 5) and *evaluation practices* (Tables 7, 8).

A. APPLICATIONS TAXONOMY AND SYSTEM PATTERNS

We group applications into eight families: (1) Medical & Pathology, (2) Information Integrity & Security (fake news, backdoor/jailbreaks), (3) Transportation & Mobility (driver monitoring), (4) Agriculture & Environment, (5) Video Understanding & Surveillance, (6) Manufacturing &

TABLE 17. Evaluation protocols by scenario and reporting checklist.

A. Choosing an evaluation protocol by scenario	
Scenario	Recommended Protocol
Open-vocabulary retrieval	Zero-shot on Flickr30k/COCO; report Recall@1/5/10, MRR, nDCG
Captioning for assistive use	Few-shot + human preference; BLEU/CIDEr + error taxonomy
Knowledge-based VQA	OK-VQA with/without retrieval; accuracy + attribution score
Doc/Chart understanding	TextVQA + visualization literacy suite [72]; EM/F1 + human study
Instruction following	MME/MMBench/MMMU; EM/F1 + pairwise preference
Safety-critical domain	Paired expert evaluation; calibration (ECE), abstention metrics
B. Result reporting checklist	
Inputs & Decoding: resolution, OCR, prompt, shots, temperature/beam, max tokens.	
Protocol: zero/few/fine-tuned; in-context examples; split names; contamination guard.	
Metrics: task (BLEU/CIDEr, Recall@k, VQA acc.) + reliability (ECE, robustness drop).	
Human eval: rubric, raters (#), expertise, agreement, cost, instructions.	
Compute: hardware, throughput/latency, memory, energy proxy if available.	
Data provenance: sources, filters, licenses, de-duplication.	
Safety: jailbreak tests used, refusal/abstention rate, red-team summary.	

Industrial QA, (7) Remote Sensing & Cartography and (8) Multimodal Agents & GUI/Docs. For each, we indicate representative datasets, core tasks, capability metrics, reliability add-ons and common system choices (Two-Tower/Two-Leg/One-Tower/Encoder–Decoder; contrastive vs. ITM/MLM vs. generative/instruction), referencing Sections III and V.

B. APPLICATION CARDS

For each domain below, we follow a compact template that covers data, tasks, metrics, reliability considerations, system patterns and open gaps. This keeps the narrative aligned with Sections V – VI while avoiding redundancy.

1) MEDICAL & PATHOLOGY

Data include CCBench [48], multi-dataset pathology in [78] and anomaly sets [45]. Baseline tasks include diagnostic classification, anomaly detection, retrieval and VQA. Throughput metrics typically include AUC/accuracy, VQA accuracy, Recall@k and agreement with the expert. Reliability focuses on calibration and abstention, domain shift with scanners and stains and explainability. System trends typically intertwine Two-Leg or Encoder–Decoder stacks with adapters, multitask generative learning with ITM/MLM and instruction tuning. Open challenges include robustness with rare.

2) INFORMATION INTEGRITY & SECURITY

Data come from IMFND for multimodal fake news detection [14] and from backdoor or jailbreak suites [15]. Top tasks

are fake-news classification and attack detection or defense. Measurements are accuracy/F1, attack success rate and evidence precision at top-k. Reliability is based on adversarial images and prompts and on attribution quality. Systems typically use instruction-tuned Two-Leg architecture with retrieval augmentation. Open gaps include cross-platform generalization and standardized red-team batteries.

3) TRANSPORTATION & MOBILITY

Data comprises the 100-Driver and corresponding DDC sets [68]. Tasks include driver state or action classification and incident captioning. Metrics include accuracy/recall/F1 with latency and FPS as a secondary set of metrics. Reliability emphasizes low-light and motion-blur stressing tests and on-vehicle compute constraints. System trends include Encoder–Decoder stacks with skill activation (segmentation or pose) and server offloading when necessary. Open challenges include privacy-preserved testing and long-horizon reasoning.

4) AGRICULTURE & ENVIRONMENT

Examples of representative data are E-CLIP fruit [50] and Eco-VLMs [66]. Tasks revolve around recognition and detection with classification of fruit freshness as the focus. F1, mAP@0.5 and FPS are common metrics. Reliability focuses on weather and illumination distribution shift and variation due to cultivar. Systems typically comprise Two-Tower models with PEFT prompts and domain augments. Open gaps are associated with.

5) VIDEO UNDERSTANDING & SURVEILLANCE

Examples of representative data are E-CLIP fruit [50] and Eco-VLMs [66]. Tasks revolve around recognition and detection with classification of fruit freshness as the focus. F1, mAP@0.5 and FPS are common metrics. Reliability focuses on weather and illumination distribution shift and variation due to cultivar. Systems typically comprise Two-Tower models with PEFT prompts and domain augments. Open gaps concern long-context memory and incident grounding.

6) MANUFACTURING & INDUSTRIAL QA

Data consist of construction scenarios [83], ChipVQA [16], wafer inspection [86] and DefectGLM [90]. Tasks encompass defect detection, plan generation and VQA. Metrics focus on AUROC/AUPRC, success in tasks and Recall@k. Reliability covers line, lot and domain shift, explainability and latency. Patterns prefer Two-Leg or joint stacks with adapters and few- or zero.

7) REMOTE SENSING & CARTOGRAPHY

Data draw from RS adapters [26], MMSHIP [92] and MapTrain/MapReader [46]. Tasks cover classification, segmentation, ship detection and map QA. Metrics include accuracy, IoU and zero-shot transfer. Reliability focuses on cross-sensor and cross-country generalization and geo-bias. System patterns combine contrastive encoders with adapters

TABLE 18. Summary of applications of visual-language models across domains.

Application Domain	Study / Framework	Key Contribution / Findings
Medical Imaging and Pathology	[48] – CCBench	Evaluated LLMs and LVLMs in cervical cytology; GPT-4 and Gemini performed well but remained below expert level.
	[78]	Proposed unified VL framework for cross-modal retrieval and patch classification; outperformed CLIP.
	[79]	Improved Ewing’s Sarcoma diagnosis using VLS within MIL; reduced parameters vs. ImageNet-based methods.
	[45]	Introduced multi-level adaptation to CLIP for anomaly detection; achieved SOTA AUC.
Fake News Detection	[14] – IMFND	LVLMs performed comparably to BERT/CLIP; IMFND framework significantly improved performance using modality-guided prompts.
Backdoor Attacks	[15] VL-Trojan	Introduced backdoor attack on VLMs using image/text triggers; showed robust ASR performance under multimodal instruction tuning.
Distracted Driving Classification	[68] – DDC-Chat	LLaVA-Plus based model using reasoning chains and instruction tuning; achieved SOTA on 100-Driver dataset in zero/few-shot settings.
Fruit Detection	[50] – E-CLIP	Multi-modal fruit recognition framework with high performance (F1=0.752, mAP@0.5=0.791) and 54.82 FPS in real-time.
Video Action Recognition	[80] – Watermark Prompts	Visual-only prompting (imperceptible watermarks/trademarks); surpassed conventional fine-tuning.
	[81]	Evaluated multi-modal VLMs on action recognition datasets; highlighted benefits of text-visual integration.
	[82] – TrafficVLM	Dense video captioning for surveillance; top 3 in AI City Challenge.
	[19] – Vid2Seq	Dense event captioning using narrated videos; SOTA on benchmarks, strong generalization.
Manufacturing	[83]	Construction scene VLM with attention & beam search; semantic analysis of complex scenes.
	[27] – Diagnostics-LLaVA	Automotive diagnostics VLM; better fault detection through fine-tuning.
	[84] – Ground4Act	Robotic grasping/planning with DQN; outperformed baselines.
	[85] – ReplanVLM	Visual planning with error correction; improved task completion in open-world robotics.
	[16] – ChipVQA	142-question benchmark for VLMs in chip design.
	[86] – Prompt-based Wafer Defect Inspection	Used Nemotron VLM; zero-shot inspection without annotated datasets.
	[87] – FontCLIP	CLIP with typographic supervision; multilingual/cross-lingual font retrieval.
	[88] – OpenECAD	Editable sketch and 3D command generation from CAD images.
	[89]	Bridge inspection via in-context VLM; accurate few-shot captioning.
	[90] – DefectGLM	36-type wafer defect classification using large-scale VLM with visual adaptation.
Remote Sensing	[91] – NavBLIP	UAV navigation using VLM with NIMFE; real-time performance in aerial inspection.
	[26] – CLIP-MoA	Adapter-based domain adaptation for CLIP; improved RS classification under few/zero-shot settings.
	[92] – Popeye	Ship detection via hybrid encoders + segmentation; robust in complex scenes.
	[93]	SAR recognition via diffusion + domain adaptation; boosted zero-shot SAR recognition.
Multimodal Agents / Assistants	[46] – MapReader	VLM fine-tuned for cartography; used MapTrain dataset for QA and description tasks.
	[94] – CogAgent	18B-param GUI navigation model with high-res input support; SOTA on GUI understanding tasks.

and instruction adaptation. Open gaps include license and provenance tracking and consistent geo-splits.

8) MULTIMODAL AGENTS & GUI/DOCS

These data comprise GUI and text-rich VQA collections from CogAgent [94]. Tasks are navigating the GUI and QA through screenshots or documents. Measurements specify success of tasks, steps-to-goal and VQA

precision. Usability indicates reliability, refusal and abstention behavior and latency. System trends employ high-definition encoders, OCR-informed prompts and decoder-based LLMs. Open gaps involve UI change robustness and accessibility measures.

VIII. CHALLENGES AND OPEN PROBLEMS

Applications in Section VII highlight both promise and pitfalls: today’s VLMs excel on curated leaderboards yet

often falter under distribution shift, noisy inputs or adversarial prompts (cf. Tables 7, 8). We distill the landscape of open problems into three broad themes: *Data & Bias*, *Evaluation & Reliability*, and *Deployment & Human Factors*.

A. DATA & BIAS

1) WHAT WE KNOW

Training corpora scraped from the web encode cultural and demographic skew, harmful content, and inconsistent licenses [47]. Domain-specific datasets (e.g., medical, remote sensing) are often narrow or uneven, while sensitive or copyrighted material may inadvertently leak into model outputs. These issues propagate across both text and vision channels.

2) OPEN PROBLEMS

How to ensure multilingual and geo-diverse coverage; how to conduct subgroup-aware audits; and how to track provenance and protect privacy at scale.

3) DIRECTIONS

Introduce standardized data cards and de-duplication protocols; adopt stratified dataset splits over geography and time; report subgroup metrics; and combine provenance logging, PII scrubbing, and license-aware filtering with human-in-the-loop data curation.

B. EVALUATION & RELIABILITY

1) WHAT WE KNOW

Current benchmarks are fragmented and task-specific, making cross-study comparisons difficult [95]. VLMs can hallucinate content not grounded in images [96], remain vulnerable to adversarial prompts [15], [75], and show weaknesses in compositional and temporal reasoning [76].

2) OPEN PROBLEMS

Establishing standardized protocols and contamination controls; designing scalable methods to verify that answers truly use visual evidence; building robust multimodal adversarial testbeds; and developing meaningful metrics for causal, temporal, and planning tasks.

3) DIRECTIONS

Protocol templates and reporting checklists; paired human preference studies with confidence intervals; retrieval-augmented generation with evidence scoring [56]; decoding strategies that balance visual and textual inputs [71]; robust fine-tuning and abstention mechanisms; and energy/FLOP reporting alongside capability metrics.

C. DEPLOYMENT & HUMAN FACTORS

1) WHAT WE KNOW

The quadratic complexity of attention constrains context length, while memory and latency limit on-device deployment. Distillation and quantization offer partial relief [49],

but compression can erode grounding. In high-stakes applications such as healthcare or industrial automation, human oversight and safe refusals are essential [94]. Training and inference at scale also carry environmental costs and exacerbate inequalities in access to models and datasets.

2) OPEN PROBLEMS

How to preserve grounding under compression; how to support multilingual and resource-efficient deployment; how to design reliable abstention and interactive correction loops; and how to balance environmental impact with equitable access.

3) DIRECTIONS

Token-aware compression and dual-modality distillation; throughput and energy reporting as standard practice; expert panels and selective prediction frameworks for oversight; user-in-the-loop repair and accessibility metrics; and sustainability-aware deployment policies that align with global resource disparities.

D. OPEN PROBLEMS AS TESTABLE RESEARCH QUESTIONS

- Can we design a benchmark that couples capability (e.g., VQA) with *mandatory* robustness, bias and calibration tracks, reported with CIs and contamination checks?
- What scalable protocols can audit whether answers *use* visual evidence rather than language priors?
- Which memory mechanisms (stateful mixing vs. retrieval) best preserve temporal consistency under edge compute budgets?
- Can LAION-scale datasets be constructed with end-to-end provenance and PII hygiene without degrading capability?
- How can we measure and improve parity across scripts and locales in OCR or text-rich tasks?

E. ETHICS, FAIRNESS AND SAFETY: FROM PRINCIPLES TO PRACTICE

Ethical deployment requires both procedural safeguards and technical measures: governance documents that specify use cases and failure modes; continuous red-teaming and user feedback loops; dataset and model cards with licensing and provenance; and standardized reporting of refusal rates, adversarial success rates, and subgroup performance gaps. Bridging the gap between benchmark proficiency and safe, reliable deployment entails: (i) diversified, well-documented data; (ii) grounding-aware training and testing; (iii) reliability checks baked into every evaluation; and (iv) runtime safeguards such as abstention, attribution, provenance logging and energy tracking. Table 19 summarizes these risks and corresponding mitigations for practitioners.

IX. FUTURE DIRECTIONS

A high-level conceptual map of ongoing and anticipated research trajectories is illustrated in Fig. 3. The diagram

TABLE 19. Risk Mitigation matrix for VLM deployment, organized by the three overarching challenge themes. Each row links challenges to concrete measurements (from Tables 7, 8) and actionable mitigations.

Challenge Theme	Measurements to Report	Mitigations / Design Choices
Data & Bias	Subgroup metrics and gaps; geo/lang coverage; dataset cards; data source and licensing audits; PII leakage tests	Stratified splits; human-in-the-loop filtering; provenance logs; license-safe filters; PII scrubbing; balanced bootstrapping; transparent documentation
Evaluation & Reliability	CHAIR/POPE grounding scores; evidence attribution precision/recall; human audits; adversarial success rate (ASR); robustness under corruptions; refusal/abstention coverage; reasoning probes; long-horizon plan quality (PG2S)	Retrieval augmentation (RAVL); re-balanced decoding (RBD); preprocessing defenses (DiffPure); robust fine-tuning (MDAPT); multimodal red-team suites; interleaved cross-attention; concept bottlenecks (DCBM); selective prediction
Deployment & Human Factors	Latency, memory, and FPS; energy per sample; throughput and cost; expert agreement (κ/α); task success rates; accessibility/usability metrics; environmental footprint	Distillation + quantization (EdgeVL, QAT); token-aware compression; PEFT; hybrid mixing (VisualRWKV-HM); expert-in-the-loop review; UI for correction; accessibility testing; sustainability-aware deployment and reporting

organizes future directions in Vision–Language Models (VLMs) into four interdependent clusters: (1) *Multimodal Instruction Tuning*, emphasizing alignment with human intent and low-supervision adaptation; (2) *Efficient Pretraining*, focusing on lightweight adapters, parameter-efficient fine-tuning, and modular transfer learning; (3) *Unified Multimodal Agents*, integrating perception, reasoning, and robotic control through shared architectures; and (4) *Open-Source and Democratization*, advancing transparency, reproducibility, and equitable access through open benchmarks and public model releases.

We outline a forward-looking research agenda that is *goal-driven* (balancing capability and reliability), *data-aware* (building on robust supervision regimes; cf. Table 5), and *evaluation-first* (anchored in metrics from Table 7 and robustness dimensions in Table 8). The agenda is organized into six axes, each pointing to specific challenges and opportunities that can guide the next phase of multimodal research.

A. DATA-CENTRIC MULTIMODALITY AND PROVENANCE

Future VLMs will be no better than their training data. A key challenge is breaking free of scale by itself towards more representative, auditable, and provenance-aware corpora. This means curating supervision with express geographic and linguistic coverage as well as shifting away from caption-only datasets towards *evidence-linked pairs* where images are paired with citations or sources. These sorts of practices would make for stronger attribution, allow faithful grounding on a larger scale, and make downstream auditing more feasible. Data cards, machine-readable metadata, and provenance-by-design must become common currency in order to facilitate both accountability as well as safe redistribution.

B. OBJECTIVE DESIGN, INSTRUCTION TUNING AND ALIGNMENT

The objective-oriented design necessitates direct coding of principles for grounding, reliability, and efficiency.

Fine-tuning will penalize hallucination and will strengthen diligent grounding, and instruction tuning can be complemented by calibrated abstention towards promoting selective prediction as well as evidence scoring. Trends such as retrieval-augmented generation, structure-aware attention mechanisms (e.g., Reformer [97]), *state space model (SSM)*-inspired visual–language backbones that reduce or replace quadratic attention, and vision-informed mixture-of-experts provide promising directions towards combining efficiency with reliability. Together, they push progress towards models that not only reach raw performance but that can be explained and trusted under decision-critical scenarios.

C. ARCHITECTURES AND SYSTEMS: TOWARD GENERALIST AGENTS

Architectural innovation will be required in transitioning towards more generalist multimodal agents from task-specific VLMs. These systems need to perceive, reason, as well as act across texts, images, video, GUIs, as well as robotic worlds [98]. Modular interfaces—like adapters, Q-formers, memory stacks, as well as concept bottlenecks—enable interpretability as well as swap-in/out-ability of skill. High-resolution as well as document/UI understanding remain significant frontiers that need OCR-aware encoders as well as layout-robust pipelines. In this sense, VLMs transition away from monolithic predictors towards more flexible systems that can reason complexly across numerous domains.

D. EVALUATION, SAFETY AND GOVERNANCE

Evaluation procedures need to scale on parity with capabilities in models. All capability benchmarks need strength tests as accompaniments, fairness, and calibration with explicitly noted confidence intervals as well as contamination controls. In addition to accuracy, the question of attribution becomes paramount: benchmarks need cited-visual evidence as well as decoding procedures that redistribute text and vision contributions. Wider governance needs not just transparent reporting but openly repeatable testbeds as well as publicly

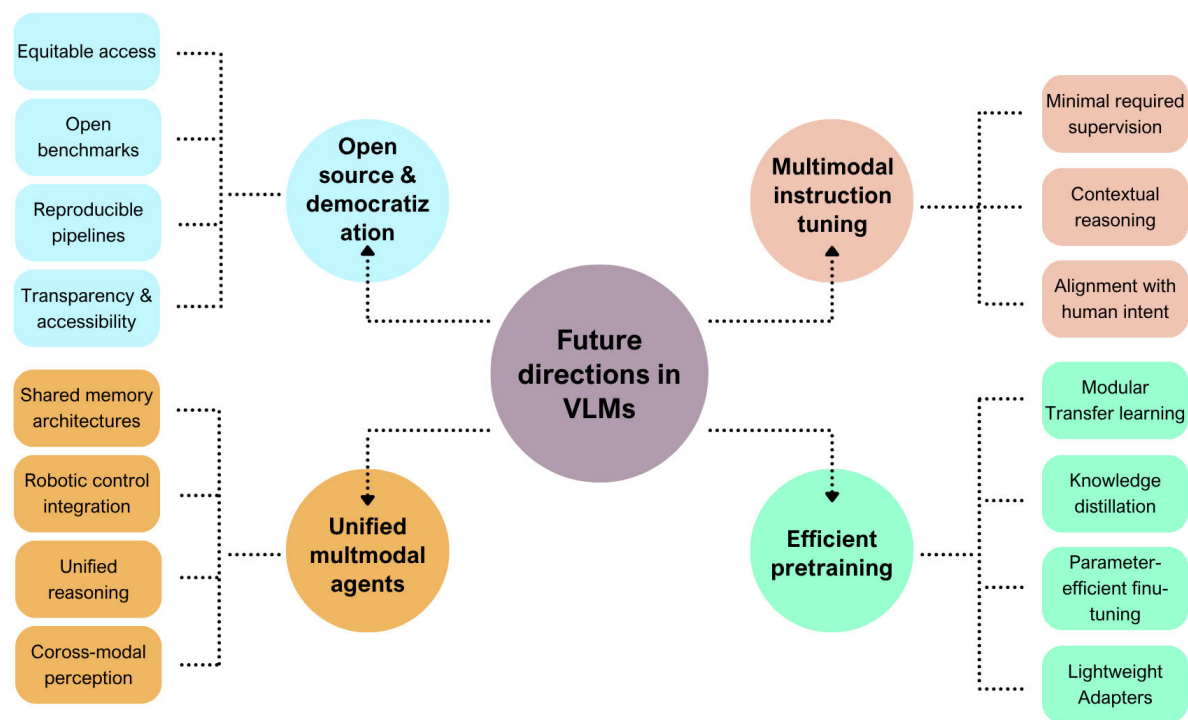


FIGURE 3. Future directions in VLMs organized along data, objectives/training, architectures/systems, evaluation/governance and deployment/efficiency.

TABLE 20. Roadmap: direction → exemplar methods → success criteria.

Direction	Exemplar Methods	Success Criteria / KPIs
Grounded instruction tuning	Visual-salience regularizers; attribution-aware losses; selective prediction	↓ CHAIR/POPE; ↑ evidence precision/recall; stable VQA acc.
Efficient pretraining	Linear/structured attention [97]; state space models (SSMs) / linear mixers (e.g., RWKV-like); MoE routing; PEFT	↓ FLOPs/memory; same or better zero-shot Recall@k / VQA
RAG with calibration	Jointly tuned retriever–generator; abstention	↑ OK-VQA; coverage–accuracy curves; calibrated refusals
Generalist agents	Modular skills; shared memory; high-res/OCR	↑ task success across domains; plan-quality metrics; robustness to UI change
Data governance	Evidence-linked pairs; data cards; license filters	Subgroup gap reports; contamination audits; reproducible evals
Edge readiness	Distillation + QAT; token-rate control	Latency/FPS/energy targets met with ≥95% task performance

accessible suites for red-teaming [99], [100]. These provisions allow gains in capability not to lead to losses in safety, reproduceability, or fair access.

E. DEPLOYMENT, EFFICIENCY AND SUSTAINABILITY

Ultimately, VLM deployment must be efficient, private, and green. Operating under constraints of latency, energy, and memory by means of distillation, quantization-aware training, and token compression will make systems practical. Privacy-preserving retrieval and provenance methods—the use of private indices, license-safe filtering, and PII scrubbing—are mandatory for safe and trustworthy uptake. Sustainability, on the other hand, necessitates carbon- as well

as cost-aware research, where training as well as inference expenses are mentioned next to performance measures. Lightweight fine-tuning approaches such as PEFT and hybrid attentional architectures are promising avenues toward optimizing accessibility, efficiency, as well as sustainability effects.

F. FROM VISION STATEMENTS TO TESTABLE QUESTIONS

The agenda seeks ultimately to transform such broad aspirations into scientifically testable questions. Can adaptation of grounded instruction diminish hallucination without losing task accuracy? Will retriever–generator training with a joint objective enhance both knowledge-driven VQA as well as

evidence attribution? To what extent can VLMs that are quantized and distilled maintain efficiency under severe deployment constraints? Can a single generalist agent exhibit parity across GUI navigational tasks, captioning tasks, as well as chart QA tasks without task-specific architectures? And how much can we get towards zeroing subgroup gaps on multilingual as well as OCR-dominated tasks? The answers towards such questions will constitute the next phase of VLM research.

The next wave of VLMs will be: (i) *grounded* (attribution-first), (ii) *efficient* (linear/PEFT/distilled), (iii) *generalist* (unified agents) and (iv) *governed* (provenance, safety, openness). This agenda converts ambitions into testable hypotheses with concrete KPIs, aligning research practice with trustworthy, deployable systems.

To consolidate the diverse challenges and opportunities identified throughout this survey, Table 20 outlines a structured research roadmap for Vision–Language Models. Each direction is paired with exemplar methodological avenues and measurable success criteria, reflecting both near-term feasibility and long-term scientific impact. The roadmap emphasizes grounded alignment, computational efficiency, reliability, and ethical governance as the key pillars guiding the next phase of multimodal AI development.

X. CONCLUSION

Visual Language Models are transforming multimodal AI through combining perception and reasoning over text and images. In this overview, we surveyed their architectures, objectives during training, evaluation frameworks and varying applications. Much has been done well, but much remains to be done: models do robustness, fairness, efficiency and explainability poorly enough, particularly in safety-critical situations. Closing these gaps needs more than algorithms but also crossdisciplinary work with experts from domains, ethicists and educators. Ahead, VLMs will likely move from static benchmarks to dynamic, in-the-world evaluation with greater focus on trustworthy deployment and domain-specific adaptation. Through foundations clarified and future research avenues marked out, our work aims to inform research and practice and assist in the responsible development of multimodal AI.

This survey organized the rapidly evolving VLM landscape along two orthogonal, high-leverage axes—*architectures* and *training objectives* (Sec. III)—and connected them to *datasets & supervision regimes* (Sec. V), *evaluation & reliability* (Sec. VI) and *applications* (Sec. VII). Our comparative matrices and cross-walk tables expose how objective choices are constrained by supervision, how model families trade efficiency for reasoning depth and how domain deployments depend on reliability add-ons beyond leaderboard metrics.

Key takeaways:

- Design levers. Two-Tower excels at scalable retrieval; Two-Leg/One-Tower strengthen token-level grounding;

encoder–decoder stacks enable instruction following (Tables 10, 4).

- Data→objective alignment. Web-scale noisy pairs favor contrastive alignment; curated/instructional corpora favor ITM/MLM/generative mixes (Table 5).
- Capability and reliability. Task scores (Table 7) are necessary but insufficient; robustness, fairness, calibration and efficiency (Table 8) determine deployability.
- Applications are data-bound. Domain performance tracks domain data and metrics; mapping apps to datasets and KPIs clarifies deployment targets (Table 14).

Despite impressive zero-/few-shot transfer, models remain sensitive to distribution shift, adversarial prompt/image attacks and dataset bias; grounded attribution is fragile and evaluation practices are inconsistent across protocols (Sec. VIII).

What practitioners should do now:

- Start with supervision fit. Choose objectives that match data regimes; favor PEFT/adapters before full finetuning (Sec. V).
- Evaluate beyond accuracy. Pair capability metrics with reliability batteries; report calibration, robustness drops, subgroup gaps and latency/energy (Sec. VI).
- Add grounding. Prefer retrieval-augmented pipelines with evidence scoring for knowledge tasks; enable selective abstention in safety-critical use (Secs. VI, VII).
- Design for deployment. Use linear/hybrid mixing, distillation and quantization for edge or cost-constrained scenarios (Secs. IV, V).

We anticipate VLMs that are *grounded* (attribution-first), *efficient* (linear/PEFT/distilled), *generalist* (unified agents spanning image/video/GUI/docs) and *governed* (provenance, fairness, safety). Bringing such vision to life entails alignment of architectural advancements with accountable data curation and reliability-minded standard assessment. These survey artifacts (taxonomies, cross-walks and checklists) aspire to function as a working guide towards VLMs of dependable, domain-ready sort.

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